SOFTWARE REQUIREMENTS SPECIFICATION

Design and Implementation of a Mobile Application for Collection of User Experience Data from Mobile Network Subscribers

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the complete functional and nonfunctional requirements for the Mobile Network Quality of Experience (QoE) Monitoring Application. The application is designed to collect real-time network performance metrics and subjective user satisfaction data from mobile network subscribers in Cameroon. This document serves as the primary reference for the development team, academic supervisors, and any stakeholders involved in evaluating or implementing the system.

1.2 Scope

The system is a native Android mobile application that runs continuously in the background on a subscriber's device. It automatically collects objective network metrics such as signal strength, network type, latency, and packet loss, while also periodically prompting users to rate their experience. The collected data is stored locally and synchronised to a remote server, where it can be accessed in anonymised form to support network quality analysis and operator decision-making.

The application is intended for deployment in Cameroon and targets subscribers of the major mobile network operators including MTN Cameroon, Orange Cameroon, Camtel, and Nexttel. It addresses a clear gap identified through requirement gathering: no existing tool combines continuous background monitoring with subjective user feedback collection in a manner accessible to everyday Cameroonian subscribers.

1.3 Definitions, Acronyms and Abbreviations

Term / Acronym	Definition
QoE	Quality of Experience — the subjective perception of service quality by the end user
QoS	Quality of Service — objective technical measurements of network performance

SRS	Software Requirements Specification
FR	Functional Requirement
Term / Acronym	**Definition**
NFR	Non-Functional Requirement
GPS	Global Positioning System — used for location tagging
RSSI	Received Signal Strength Indicator — a measure of signal power
LTE	Long-Term Evolution — commonly known as 4G mobile network technology
ART	Agence de Regulation des Telecommunications — Cameroon's telecom regulator
ITU	International Telecommunication Union
APK	Android Package Kit — the installable file format for Android apps
SDK	Software Development Kit
API	Application Programming Interface

1.4 Document Overview

This document is structured as follows: Section 2 provides an overall description of the system. Section 3 identifies the stakeholders. Sections 4 and 5 detail the functional and non-functional requirements respectively. Section 6 presents use cases. Sections 7 through 12 cover system constraints, prioritisation, data dictionary, assumptions, validation, and glossary.

2. Overall System Description

2.1 Product Perspective

The Mobile Network QoE Monitoring Application is a standalone native Android application. It operates independently on the subscriber's device without requiring any integration with the network operator's infrastructure for its core functionality. Data collected is stored locally and can be synchronised to a backend server when internet connectivity is available. The application occupies a unique position in the mobile analytics landscape by combining passive background monitoring (traditionally operator-side) with active user feedback collection (traditionally survey-based) into a single lightweight mobile tool.

2.2 Product Functions — Summary

The application performs the following high-level functions:

- Continuously monitors network quality metrics in the background
- Periodically prompts the user to rate their network experience
- Stores all collected data locally with timestamps and GPS coordinates
- Displays a real-time dashboard showing current network status
- Shows the user a personal history of their network performance
- Provides a network coverage map based on crowdsourced readings
- Detects and alerts the user to unusual network quality drops
- Synchronises data to a remote server for operator-level analysis
- Supports dual SIM device detection and per-SIM monitoring

2.3 User Classes and Characteristics

User Class	Description	Technical Level
Mobile Subscriber	Everyday Cameroonian phone users aged 18 and above who install and use the app on their personal Android device	Low, non-technical

Network Operator Analyst	Personnel at MTN, Orange, or other operators who access aggregated anonymised data from the backend	High
Developer / Administrator	The team responsible for building, deploying, and maintaining the app and its backend	High

2.4 Operating Environment

- Platform: Android 8.0 (Oreo) and above
- Devices: Entry-level Android devices (Tecno, Itel, Infinix) and mid-to-high range Android devices (Samsung A/S series, Google Pixel)
- Network: Supports 2G (EDGE), 3G, and 4G/LTE environments
- Connectivity: Must function in both online and offline conditions
- Location: Designed for use across urban, peri-urban, and rural areas of Cameroon

2.5 Design and Implementation Constraints

- The application must be developed as a native Android application to ensure full access to background services, network state APIs.
- A web application architecture is not feasible as it cannot run continuously in the background or access device network metrics
- The application must function on devices with as little as 2GB RAM
- Data transmitted to any server must be encrypted
- User location data must be anonymised before any storage or transmission

3. Stakeholders

The following stakeholders were identified through the requirement gathering phase and have a direct or indirect interest in the design, implementation, and outcomes of this system.

Stakeholder	Role / Interest	Impact on System
Mobile Subscribers	Primary end users of the app. Their feedback and usage data are the core output of the system	Define usability, acceptance, and feature requirements

Mobile network operators	Mobile network operators who benefit from the aggregated QoE data to improve network planning and customer experience	Define what data is most useful and how it should be structured
ART Cameroon	Regulatory body responsible for telecom quality standards in Cameroon	Defines regulatory boundaries for data collection and privacy
Development Team	Responsible for designing, building, testing, and deploying the application	Translate requirements into working software

4. Functional Requirements

Functional requirements define what the system must do. Each requirement is derived directly from the survey data collected from 50 mobile network subscribers across Cameroon, supplemented by reverse engineering findings, brainstorming sessions, and document analysis of ITU QoE standards. Requirements are written using the standard convention: "The system shall..."

4.1 Background Monitoring

ID	Description	Source
FR-01	The system shall run as a persistent background service that continues operating even when the application is not open by the user.	Survey Q19: Users expect the app to monitor quietly without needing to be opened
FR-02	The system shall automatically restart its background service if it is terminated by the Android operating system.	NFR dependency; continuous monitoring cannot have gaps
FR-03	The system shall collect network metrics at configurable intervals, with a default interval of every five minutes.	Brainstorming: balances data richness with battery and data consumption

4.2 Network Metric Collection

ID	Description	Source
FR-04	The system shall collect and log the current network type (2G/EDGE, 3G, 4G/LTE) at each measurement interval.	Survey Q9: Users connect on all three generations; all must be detected and recorded
FR-05	The system shall collect and log the current signal strength (RSSI/RSRP) at each measurement interval.	ITU-T QoE standards; identified as a primary metric in document analysis
FR-06	The system shall measure and log network latency (ping) at each measurement interval.	Document analysis; ITU defines latency as a core QoE metric
ID	**Description**	**Source**
FR-07	The system shall measure and log download and upload bandwidth speed at configurable intervals.	Survey Q24: Internet speed history was the most requested feature
FR-08	The system shall detect and log packet loss percentage during active data sessions.	Document analysis, ITU core QoE metric
FR-09	The system shall record the GPS location (latitude and longitude, anonymised to approximate area) with every metric reading.	Survey Q22, Majority accepted approximate location; exact coordinates must not be stored
FR-10	The system shall attach a timestamp to every data record collected.	Project specification; fundamental requirement for temporal analysis

4.3 Dual SIM Support

ID	Description	Source
FR-11	The system shall detect whether the device contains one or two active SIM cards.	Survey Q7: A significant proportion of respondents use dual SIM devices
FR-12	The system shall independently monitor and log metrics for each active SIM card separately.	Survey Q7: Dual SIM users switch between networks; both must be monitored

4.4 User Feedback Collection

ID	Description	Source
FR-13	The system shall offer two modes of user interaction: (a) Prompted Mode, where the user is periodically asked to rate their network experience, and (b) Fully Automatic Mode, where no prompts are shown and all data is collected silently.	Survey Q21: A significant portion of respondents preferred fully automatic mode with no prompts
FR-14	In Prompted Mode, the system shall send a notification asking the user to rate their current network experience on a scale of 1 to 5.	Survey Q21: Preferred prompt frequency was once per day or only when network is bad
FR-15	The system shall allow the user to select the trigger for feedback prompts: once daily, a few times per week, or only when a significant network drop is detected.	Survey Q21: Different users expressed different preferred frequencies
FR-16	The system shall allow the user to optionally write a short free-text note describing their current network experience alongside any rating submitted.	Project specification capturing user stories and narrative experience is a core project aim

FR-17	The system shall log all submitted ratings and notes together with the corresponding network metrics and timestamp at the time of submission.	Project specification linking subjective feedback to objective data is the core research value

4.5 Network Problem Detection and Alerts

ID	Description	Source
FR-18	The system shall define threshold values for network quality below which a condition is classified as a significant network drop.	Brainstorming threshold definition enables intelligent alerting
FR-19	The system shall send the user a notification when a significant and sustained network quality drop is detected.	Survey Q13: Users experience daily problems with no current automatic notification mechanism
FR-20	The system shall log the time, location, network metrics, and type of problem (dropped call, slow data, no signal, failed mobile money transaction) associated with each detected network drop event.	Survey Q14: Users reported all five problem types as frequent occurrences

4.6 Data Storage and Synchronisation

ID	Description	Source
FR-21	The system shall store all collected metric data, user ratings, and notes locally on the device in a structured database.	Project specification local storage enables offline operation

FR-22	The system shall continue collecting and storing data locally even when the device has no internet connectivity.	Survey Q16 — Users in rural and remote areas frequently report complete signal loss
FR-23	The system shall synchronise locally stored data to a remote server when internet connectivity is restored.	Project specification, server-side data enables operator-level analysis
FR-24	The system shall store a minimum of 30 days of collected data locally before overwriting the oldest records.	Brainstorming provides sufficient history for meaningful personal and research analysis

4.7 User Interface and Dashboard

ID	Description	Source
FR-25	The system shall display a real-time dashboard showing the user's current signal strength, network type, and internet speed.	Survey Q24: Users want visibility of their current network status
FR-26	The system shall display a personal history of the user's network performance over the past 7 and 30 days in a simple graph or chart format.	Survey Q24: 'Seeing my own internet speed history' was the most selected feature across all respondents
FR-27	The system shall display a network coverage map showing signal quality readings at different locations based on collected data.	Survey Q24: 'A map showing the best signal areas in my city' was the second most selected feature

FR-28	The system shall display a comparative view of network performance between the user's active SIM cards or operators, where dual SIM data is available.	Survey Q24: 'Comparison of my network vs other operators' was frequently selected by dual SIM users
ID	**Description**	**Source**
FR-29	The system shall present the user's overall network quality as a simple QoE score out of 10, calculated from a weighted combination of collected metrics and user ratings.	Feature design: makes technical data accessible and understandable to non-technical users

4.8 Privacy and Data Control

ID	Description	Source
FR-30	The system shall present a clear and plain-language privacy notice to the user before any data collection begins, explaining what is collected, how it is used, and who can access it.	Survey Q20: Trust and privacy concerns were among the top barriers to app acceptance
FR-31	The system shall allow the user to delete all locally stored data at any time from within the application settings.	Survey Q23: Multiple respondents selected 'Only me' as the preferred data access group
FR-32	The system shall allow the user to opt out of data synchronisation to the remote server while continuing to use local features.	Survey Q23: Several respondents selected 'Nobody' for data access, indicating desire for local-only use

5. Non-Functional Requirements

Non-functional requirements define the quality standards and constraints the system must meet. These requirements are equally as important as functional requirements, as they determine whether the system is practical and trustworthy enough for real-world adoption by Cameroonian subscribers.

5.1 Performance

ID	Description	Source
NFR-01	The system's background service shall consume no more than 5% of total device battery per hour during normal monitoring operation.	Survey Q20: Battery drain was the most frequently cited concern across all respondents
NFR-02	The system shall use no more than 10MB of mobile data per day for its own background data collection and synchronisation operations.	Survey Q20: Mobile data consumption was the second most cited concern
NFR-03	The application's main dashboard screen shall load and display current readings within 3 seconds of being opened.	General usability best practice
NFR-04	The system shall not cause a measurable increase in device temperature during normal background operation.	Survey Q20: Phone slowing down was a commonly cited concern indicating device strain

5.2 Usability

ID	Description	Source
NFR-05	A first-time user shall be able to install the application, complete the onboarding process, and understand the main dashboard without any external guidance or training.	Survey Q19: Many respondents said 'maybe depending on performance,' suggesting simplicity is critical to winning user trust

ID	Description	Source
NFR-06	All feedback prompts and user-facing screens shall use simple, everyday language with no technical telecommunications terminology.	Survey context: Majority of respondents are students and general users with no technical background
NFR-07	The application shall support both English and French languages, with the user able to select their preferred language during onboarding.	Survey Q4: Respondents came from both Anglophone (Buea, Bamenda) and Francophone (Yaounde, Douala, Dschang) regions
NFR-08	The feedback prompt notification shall require no more than 30 seconds of user interaction to complete a rating.	Survey Q21: Users indicated preference for minimal interruption; prompts must be quick and non-burdensome

5.3 Reliability

ID	Description	Source
NFR-09	The system's background service shall maintain an uptime of at least 99% during periods when the device is powered on and not in power-saving mode.	Project specification, continuous monitoring is a core requirement; gaps reduce data quality
NFR-10	The system shall not lose any locally stored data in the event of an application crash or unexpected device restart.	FR-21 dependency, data integrity is fundamental to research value

NFR-11	The system shall gracefully handle the absence of GPS signal by recording the last known location with a flag indicating the reading was approximate.	Survey context, users in rural and remote areas regularly lose both network and GPS signals

5.4 Security and Privacy

ID	Description	Source
NFR-12	All GPS coordinates shall be rounded to an accuracy of approximately 500 metres before storage or transmission, ensuring no exact user location is ever recorded.	Survey Q22: Most users who accepted location collection explicitly requested approximate location only
NFR-13	All data transmitted between the application and the remote server shall be encrypted using HTTPS with TLS 1.2 or above.	General security best practice and data privacy requirement
NFR-14	No personally identifiable information — including name, phone number, or device IMEI — shall be collected, stored, or transmitted by the application.	Survey Q20 and Q23 : Privacy concerns were a top barrier; several users selected 'Nobody' for data access
NFR-15	The application shall request only the minimum Android permissions necessary for its operation, and shall explain to the user in plain language why each permission is required.	Survey Q20: Trust in unknown apps was a frequently cited concern

5.5 Compatibility

ID	Description	Source

NFR-16	The application shall be compatible with Android version 8.0 (Oreo, API level 26) and all subsequent versions.	Survey Q5: Device range across respondents spans from entry-level to high-end Android; Android 8.0 covers the vast majority
NFR-17	The application shall function correctly on entrylevel Android devices including Tecno, Itel, and Infinix smartphones with a minimum of 2GB RAM.	Survey Q5: Entry-level Android devices represented a significant proportion of respondents' devices
NFR-18	The application shall detect and handle both single SIM and dual SIM device configurations without errors.	Survey Q7: A significant proportion of respondents use dual SIM devices

5.6 Maintainability

ID	Description	Source
NFR-19	The application codebase shall be structured using a recognised architectural pattern (such as MVVM) to ensure it can be maintained and extended by future developers.	Academic and engineering best practice
NFR-20	The application shall include logging of internal errors and exceptions to facilitate debugging without exposing sensitive data.	Development best practice

6. Use Cases

The following use cases describe key interactions between the system and its primary users. Each use case is derived from the survey data and project specifications.

UC-01: Subscriber Installs and Onboards the App

Field	Description
Actor	Mobile Subscriber
Precondition	User has downloaded the APK on an Android device running version 8.0 or above
Main Flow	1. User opens the app for the first time 2. App displays a plainlanguage privacy notice 3. User reads and accepts the privacy notice 4. App requests necessary Android permissions (location, network state, notifications) 5. User grants permissions 6. App presents language selection (English or French) 7. App presents monitoring mode selection (Prompted or Fully Automatic) 8. App begins background monitoring and displays the dashboard
Alternative Flow	If user denies location permission, app informs them that coverage map features will be unavailable and continues with other features only
Postcondition	Background service is running; data collection has begun
Related FRs	FR-01, FR-09, FR-13, FR-30

UC-02: Background Network Monitoring

Field	Description
Actor	System (automated — no user action required)
Precondition	Background service is active; device is powered on
Main Flow	1. Every five minutes, system collects signal strength, network type, latency, bandwidth, and packet loss 2. System records approximate GPS location 3. System attaches a timestamp to the reading 4. System stores the reading in the local database 5. If internet is available, system queues data for server synchronisation

Alternative Flow	If GPS is unavailable, last known location is recorded with an approximation flag. If internet is unavailable, data is stored locally only and synchronised when connectivity is restored
Field	**Description**
Postcondition	A new timestamped and location-tagged metric record is saved to the local database
Related FRs	FR-01 through FR-10, FR-21 through FR-23

UC-03: User Receives and Responds to a Feedback Prompt

Field	Description
Actor	Mobile Subscriber
Precondition	User has selected Prompted Mode; prompt trigger condition has been met (e.g. once daily, or a network drop detected)
Main Flow	1. System sends a notification to the user's device 2. User taps the notification 3. App opens a simple one-screen prompt asking the user to rate their current network experience from 1 to 5 4. User selects a rating 5. Optionally, user writes a short note about their experience 6. User submits the response 7. System logs the rating, note, current metrics, and timestamp together
Alternative Flow	If user ignores the notification, no data is logged and the next prompt is scheduled normally
Postcondition	A combined subjective and objective QoE record is saved
Related FRs	FR-13 through FR-17

UC-04: User Views Dashboard and Network History

Field	Description
Actor	Mobile Subscriber
Precondition	At least one metric reading has been collected and stored
Main Flow	1. User opens the application 2. Dashboard displays current signal strength, network type, internet speed, and QoE score 3. User navigates to history view
Field	**Description**
	4. App displays a graph of network performance over the past 7 or 30 days 5. User navigates to the coverage map 6. Map displays colour-coded signal quality readings at recorded locations
Alternative Flow	If fewer than five readings have been collected, the app displays a message indicating that more data is needed for the history and map views
Postcondition	User has viewed their personal network performance data
Related FRs	FR-25, FR-26, FR-27, FR-28, FR-29

7. System Constraints

7.1 Platform Constraint

The application must be developed as a native Android application. A web application, Ionic/Cordova hybrid application, or iOS application cannot meet the core requirements of continuous background operation and full network state API access. React Native or Flutter may be considered but only if the development team can verify that all background service and network switching APIs are fully accessible via native plugins.

7.2 Regulatory Constraint

All data collection must comply with applicable Cameroonian laws on data protection and telecommunications. User consent must be obtained explicitly before any data collection begins. No data that could personally identify a subscriber may be collected, stored, or transmitted.

7.3 Device Constraint

The application must operate on devices with as little as 2GB RAM and 16GB internal storage. It must not require a data connection to perform its core monitoring functions. Battery consumption must be optimised to avoid user uninstallation, as identified as the primary concern in the requirement gathering survey.

7.4 Network Constraint

The application must handle 2G, 3G, and 4G/LTE network conditions gracefully. It must not assume that a reliable data connection is always available, as a significant portion of survey respondents reported experiencing complete signal loss in rural and remote areas.

7.5 Time and Resource Constraint

This project is being developed within the scope and timeline of a university research project. Features classified as 'Could Have' or 'Won't Have' in the MoSCoW prioritisation (Section 8) are explicitly out of scope for the initial version and may be addressed in future iterations.

8. Requirement Prioritisation (MoSCoW Method)

Requirements have been prioritised using the MoSCoW method, where Must Have represents the minimum viable product, Should Have represents important but non-critical features, Could Have represents desirable enhancements, and Won't Have represents features explicitly out of scope for this version.

8.1 Must Have

The following requirements are essential. The application cannot be considered functional without them.

Requirement ID(s)	Feature
FR-01, FR-02, FR-03	Background service with automatic restart
FR-04, FR-05, FR-06, FR-07	Collection of core network metrics (network type, signal, latency, speed)
FR-09, FR-10	Location tagging and timestamping of all readings
FR-13, FR-14, FR-15	User feedback prompt system with mode selection
FR-16, FR-17	Free-text experience notes linked to metric readings
FR-21, FR-22	Local offline data storage
FR-30	Privacy notice and user consent
NFR-01, NFR-02	Battery and data consumption limits
NFR-05, NFR-06	Simple, non-technical user interface
NFR-12, NFR-13, NFR14	Location anonymisation and data security
NFR-16, NFR-17	Android 8.0+ and entry-level device compatibility

8.2 Should Have

These features significantly improve the value and usability of the application but are not strictly required for a minimum viable version.

Requirement ID(s)	Feature
FR-11, FR-12	Dual SIM detection and independent per-SIM monitoring
Requirement ID(s)	**Feature**

FR-18, FR-19, FR-20	Automatic anomaly detection and network drop alerts
FR-23, FR-24	Server synchronisation and 30-day local data retention
FR-25, FR-26	Real-time dashboard and personal history graph
FR-29	QoE score calculation and display
NFR-07	Bilingual support (English and French)
NFR-09, NFR-10	99% background service uptime and crash-safe data storage

8.3 Could Have

These are desirable enhancements that would strengthen the application, particularly in a competition context, but can be deferred to a later version.

Requirement ID(s)	Feature
FR-27	Network coverage heatmap
FR-28	Operator comparison view for dual SIM users
FR-31, FR-32	User data deletion and opt-out of server sync
—	Weekly automated personal network performance summary notification
—	Reward or incentive system for sustained user engagement

8.4 Won't Have (This Version)

Feature	Reason for Exclusion

Automatic network switching between SIMs	Technically complex and unreliable on most Android versions; deferred to future AIpowered version
iOS version	Apple's restrictions make background monitoring and network API access infeasible within project scope
Real-time operator-facing analytics dashboard	Requires backend infrastructure beyond project scope; data can be exported instead
Feature	**Reason for Exclusion**
In-app rewards or airtime disbursement	Requires operator partnerships and payment integration beyond scope

9. Data Dictionary

The following table defines the key data entities collected and stored by the system.

Field Name	Data Type	Unit	Description
reading_id	String (UUID)	—	Unique identifier for each metric reading
timestamp	DateTime	ISO 8601	Date and time the reading was collected
latitude	Float (rounded)	Degrees	Approximate latitude, rounded to 500m accuracy
longitude	Float (rounded)	Degrees	Approximate longitude, rounded to 500m accuracy

network_type	Enum	—	Network generation: 2G, 3G, or 4G/LTE
operator_name	String	—	Name of the active network operator for this SIM
sim_slot	Integer	—	SIM slot identifier: 1 or 2 (for dual SIM devices)
signal_strength	Integer	dBm	Received signal strength in decibelsmilliwatts
latency	Integer	ms	Measured round-trip delay in milliseconds
download_speed	Float	Mbps	Measured download bandwidth speed
upload_speed	Float	Mbps	Measured upload bandwidth speed
packet_loss	Float	Percentage	Percentage of data packets lost during measurement
qoe_score	Float	0.0 – 10.0	Calculated overall Quality of Experience score
user_rating	Integer	1 – 5	Subjective satisfaction rating submitted by user
user_note	String	—	Optional free-text note submitted by user with rating
is_drop_event	Boolean	—	Flag indicating whether this reading was triggered by a detected network drop
synced_to_server	Boolean	—	Flag indicating whether this record has been synchronised to the remote server

10. Assumptions and Dependencies

10.1 Assumptions

- It is assumed that the target users own Android smartphones running Android 8.0 or above, which is consistent with the device profile of survey respondents.
- It is assumed that most users have access to mobile data, even if intermittently, allowing for periodic server synchronisation.
- It is assumed that users will grant the necessary Android permissions (network state, location, notifications) upon understanding the purpose of the application.
- It is assumed that network operators in Cameroon will find value in the anonymised crowdsourced data produced by the system, even if no formal partnership is established at this stage.

10.2 Dependencies

- Android Telephony API, required for reading network type, signal strength, and SIM information.
- Android Location API, required for GPS coordinate collection.
- Android WorkManager or Foreground Service API, required for reliable background processing.
- A remote server with a REST API endpoint, required for data synchronisation (FR-23).
- Google Maps SDK or OpenStreetMap, required for the network coverage map feature (FR27).
- A local SQLite or Room database, required for offline data storage (FR-21).

11. Validation Summary

Requirements were validated through the following methods to ensure they are complete, consistent, and aligned with stakeholder needs.

Validation Method	Outcome
Survey analysis (190 respondents)	All functional requirements in Sections 4.1 through 4.8 are directly traceable to one or more survey questions or responses

Cross-checking for contradictions	No functional requirements were found to contradict each other. One ambiguity was resolved: users who prefer fully automatic mode (FR-13b) are not shown any prompts, while all other data collection continues normally
Reverse engineering review	Confirmed that no existing app (Speedtest, nPerf, OpenSignal) fulfils FR-01, FR-13, FR-16, and FR-17 simultaneously, validating the novelty and need for this system
ITU standard alignment	FR-04 through FR-08 align with the six core QoE metrics defined in ITU-T P.10/G.100
Invalid response removal	One response from a respondent located outside Cameroon (Toronto) was identified and removed during data cleaning, ensuring all requirements reflect the Cameroonian user context

All requirements listed in this document are classified as either confirmed or conditionally confirmed pending technical feasibility testing during the design and implementation phase.

12. Glossary

Term	Definition
Background Service	An Android component that performs operations without a user interface, continuing to run even when the app is not open
Crowdsourcing	The collection of data from a large group of users, each contributing small amounts, to build a comprehensive dataset
Foreground Service	An Android background service that displays a persistent notification to inform the user it is running, required by Android for long-running background tasks

Heatmap	A visual representation of data where geographic areas are colourcoded based on values — in this case, network signal strength
Jitter	Variation in the time between data packets arriving, which causes instability in real-time services like calls and video streaming
Latency	The time delay between sending a network request and receiving a response, measured in milliseconds
Packet Loss	The percentage of data packets transmitted over a network that fail to reach their destination
QoE Score	A single numerical value combining multiple objective and subjective measurements to represent overall user satisfaction with network quality
RSSI	Received Signal Strength Indicator — a measurement of how well a device can receive a signal from a base station, expressed in dBm
Room Database	Android's recommended local database library, built on top of SQLite, used for persistent on-device data storage
SIM Slot	A physical slot in a mobile device that holds a SIM card. Dual SIM devices have two slots and can operate on two networks simultaneously
Threshold	A defined boundary value below which a measurement is classified as poor or triggering an alert
WorkManager	An Android API for scheduling deferrable and guaranteed background work that remains pending even if the app exits or the device restarts